Simple Linear Regression: The Setup

We want to model the relationship between a continuous predictor X and a continuous response Y. We assume the true relationship is a straight line, corrupted by random noise.

The Model

For each observation i from 1 to n:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

- β_0: The true population intercept.
- β_1: The true population slope.
- ϵ_i: The random error (residual) for observation i.

Simple Linear Regression: The Setup

Assumptions

To do statistical inference, we must make strict assumptions about the noise ϵ_i:

1. **Mean Zero:** $E[\epsilon_i] = 0$ (the line is centered).
2. **Homoscedasticity:** $Var(\epsilon_i) = \sigma^2$ (variance is constant across all X).
3. **Independence:** $Cov(\epsilon_i, \epsilon_j) = 0$ (no autocorrelation).
4. **Normality:** $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ (required for t-tests and F-tests).

Linear Regression

How do we mathematically fit a line to data? Let's treat it as a formal probability problem. We want to model a response Y given a single predictor X.

The Statistical Model

We assume the true relationship is a straight line, but every observation is corrupted by random noise:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Crucially, we assume this noise is Normally distributed: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$

The Conditional Distribution

Because the line $\beta_0 + \beta_1 x_i$ is a fixed, deterministic value for any specific x_i, the randomness comes entirely from ϵ_i. Therefore, the data follows a Normal distribution centered on the line:

$$y_i \mid x_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$$

The Line Represents an Expected Response

The regression line is not intended to pass through every observed pair (x_i, y_i). Responses vary around the line because of the random error ϵ_i.

The Population Relationship

Since the error has conditional mean zero, $E[\epsilon_i \mid X_i = x] = 0$, the line gives the expected response at each predictor value:

$$E[Y_i \mid X_i = x] = \beta_0 + \beta_1 x.$$

We use the data to estimate this average relationship. The fitted value at x estimates the mean response, not the exact outcome for every individual with that x.

The Likelihood of the Line

With n independent observations, we can construct the Likelihood Function to find the most probable values for our unknown parameters $(\beta_0, \beta_1, \sigma^2)$.

The Joint Likelihood

We multiply the Normal probability densities for all n data points:

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}\right)$$

The Log-Likelihood

Taking the natural log turns the product into a sum, pulling the exponent down:

$$\ell(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2$$

Method of Least squares

We want to find the exact values of β_0 and β_1 that maximize the log-likelihood.

Look closely at the structure of the equation:

$$\ell(\beta_0, \beta_1, \sigma^2) = \text{Constant} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Least Squares

Because the summation term is being **subtracted**, the only way to maximize the overall probability is to **minimize** that summation term!

This is called the **Method of Least Squares**. By taking the partial derivatives of this sum and setting them to zero, we derive our Maximum Likelihood Estimators.

The Method of Least Squares

Our goal is to find the values of β_0 and β_1 that minimize the sum of squared errors. Let's define this sum as a function S:

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

To find the minimum, we take the partial derivative with respect to each parameter and set them exactly to zero.

1. **With respect to the intercept (β_0):**

$$\frac{\partial S}{\partial \beta_0} = \sum_{i=1}^{n} 2(y_i - \beta_0 - \beta_1 x_i)(-1) = 0$$

2. **With respect to the slope (β_1):**

$$\frac{\partial S}{\partial \beta_1} = \sum_{i=1}^{n} 2(y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0$$

Solving for the Intercept ($\hat{\beta}_0$)

Dividing out the -2 and distributing the summation gives us our first **Normal Equation**:

$$\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \beta_0 - \beta_1 \sum_{i=1}^{n} x_i = 0$$

$$\sum_{i=1}^{n} y_i - n\beta_0 - \beta_1 \sum_{i=1}^{n} x_i = 0$$

$$\bar{y} - \beta_0 - \beta_1 \bar{x} = 0 \implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Note: *This proves that the regression line pass perfectly through the center of mass* $(\bar{x}, \bar{y})$ *of the data!*

Solving for the Slope ($\hat{\beta}_1$)

Similarly, we have : $\sum_{i=1}^{n} x_i(y_i - (\bar{y} - \hat{\beta}_1\bar{x}) - \hat{\beta}_1 x_i) = 0$

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^{n} x_i(x_i - \bar{x}) = 0$$

$$\hat{\beta}_1 \sum_{i=1}^{n} x_i(x_i - \bar{x}) = \sum_{i=1}^{n} x_i(y_i - \bar{y})$$

The Final Estimator

Using $\sum x_i(x_i - \bar{x}) = \sum(x_i - \bar{x})^2)$ and $\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$, we get $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Least Squares as an Orthogonal Projection

In the usual scatterplot, OLS minimizes squared **vertical residuals** $e_i = y_i - \hat{y}_i$, not perpendicular point-to-line distances. Minimizing perpendicular distances would define a different fitting method.
Write the model for all observations as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$. OLS chooses

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2, \qquad \hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Projection in Response Space

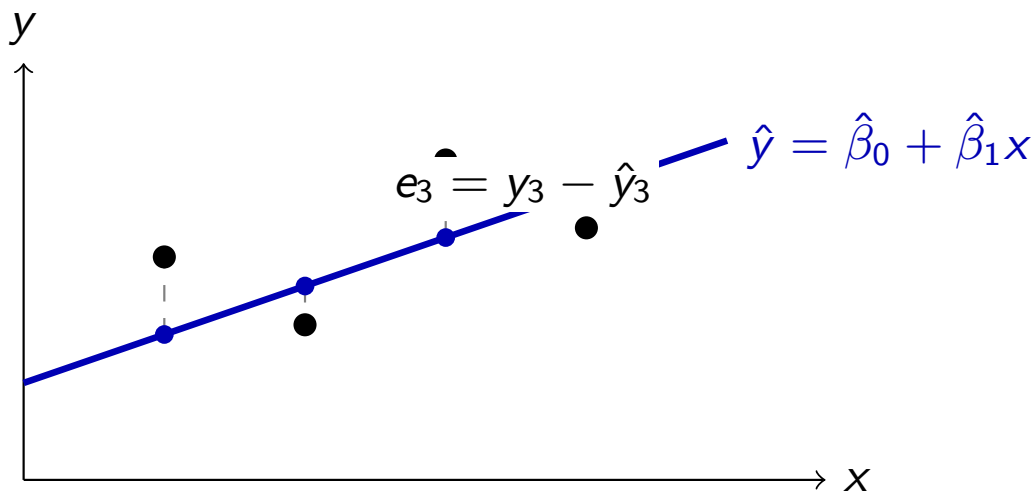
In $\mathbb{R}^n$, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$. Thus the residual vector is perpendicular to every column of $\mathbf{X}$:

$$\mathbf{X}^T(\mathbf{y} - \hat{\mathbf{y}}) = 0 \quad \Longleftrightarrow \quad \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}.$$

If $\mathbf{X}$ has full column rank, $\hat{\mathbf{y}} = \mathbf{P}_X\mathbf{y}$, where $\mathbf{P}_X = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$.

Visualizing OLS Residuals and Projection

In the scatterplot, OLS measures each residual vertically at the observed predictor value: Black dots are observations, blue dots are fitted values, and dashed segments show residuals.



The fit minimizes $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. In response space, $\hat{\mathbf{y}} = \mathbf{P}_X\mathbf{y}$ and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to the model space: $\mathbf{X}^T\mathbf{e} = 0$. This orthogonality lives in $\mathbb{R}^n$, not as perpendicular point-to-line distances in the scatterplot.

Extension to Multiple Predictors

With p predictors, include the intercept as the first column of the design matrix:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}.$$

Least Squares and Projection

As before, OLS minimizes the squared residual length and projects $\mathbf{y}$ onto the column space of $\mathbf{X}$:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2, \qquad \hat{\mathbf{y}} = \mathbf{P}_X \mathbf{y}.$$

Setting the derivative to zero gives the normal equations. If $\mathbf{X}$ has full column rank,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

With one predictor, the rows of $\mathbf{X}$ are $(1, x_i)$, recovering simple linear regression.

The Distribution of our Estimates

We know our estimators $(\hat{\beta}_0, \hat{\beta}_1)$ are calculated from our data. But since our data (y_i) contains random noise, our estimates are also **random variables**! What is their exact distribution?

Rewriting the Slope

Let's look closely at our formula for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

By expanding the numerator and recognizing that $\sum(x_i - \bar{x})\bar{y} = 0$, we can rewrite this strictly as a weighted sum of our response variables y_i:

$$\hat{\beta}_1 = \sum_{i=1}^{n}\left[\frac{x_i - \bar{x}}{\sum(x_i - \bar{x})^2}\right] y_i = \sum_{i=1}^{n} w_i y_i$$

Note: *The weights w_i are fixed constants, because we assume our X values are fixed and known.*

The Distribution of $\hat{\beta}_1$

We just proved that $\hat{\beta}_1 = \sum w_i y_i$.
Recall our core assumption: $y_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$.

We also know that **any linear combination of independent Normal variables is also Normal.** Therefore, $\hat{\beta}_1$ is perfectly Normal!

Mean and Variance

- **Expected Value:** $E[\hat{\beta}_1] = \sum w_i E[y_i] = \cdots = \beta_1 \sum_{i=1}^{n} \left[\frac{(x_i - \bar{x})x_i}{\sum (x_i - \bar{x})^2} \right] = \beta_1$
  (This proves OLS is an unbiased estimator!)
- **Variance:** Since the y_i are independent, the variance of the sum is the sum of the variances. We just square the constants:

$$Var(\hat{\beta}_1) = \sum w_i^2 Var(y_i) = \sum \left(\frac{x_i - \bar{x}}{\sum (x_i - \bar{x})^2} \right)^2 \sigma^2 = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

The Distribution of $\hat{\beta}_0$

We can apply the exact same logic to the intercept: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.
Because $\bar{y}$ is a linear combination of y_i, and $\hat{\beta}_1$ is a linear combination of y_i, their difference $\hat{\beta}_0$ is also just a linear combination of Normal variables!

The Final Distributions

Both of our OLS estimators follow exact Normal distributions:

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right)$$

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right]\right)$$

Why does this matter?

Because we now know the exact probability distributions, we can construct Confidence Intervals and perform Hypothesis Tests (like the t-test) to determine if our predictors are statistically significant!

Estimating the True Noise Variance (σ^2)

To perform any statistical test, we need to know the variance of the underlying noise (σ^2). But the true errors ($\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$) are invisible to us.

The Residuals

We can only observe the **residuals** from our fitted line:

$$\hat{e}_i = y_i - \hat{y}_i$$

Because the true errors,β_0 and β_1 are Normal, the residuals are also Normally distributed. We use the sum of their squares (SS_{res}) to estimate the true noise variance.

$$\hat{\sigma}^2 = \frac{SS_{res}}{n-2} = \frac{\sum(y_i - \hat{y}_i)^2}{n-2}$$

Why $n-2$? We started with n independent data points. But we had to estimate two parameters ($\hat{\beta}_0$ and $\hat{\beta}_1$) to calculate the predicted values $\hat{y}_i$. Because of this we are only left with $n-2$.

Confidence Intervals for the Regression Coefficients

Under independent Normal errors with constant variance, replacing σ by its estimate gives t-based intervals. Define

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad s = \sqrt{\frac{SS_{res}}{n-2}}, \qquad t^* = t_{n-2,\,1-\alpha/2}.$$

Here, t^* is the $1 - \alpha/2$ quantile of the t distribution with $n - 2$ degrees of freedom.

Two-sided $(1 - \alpha)100\%$ Confidence Intervals

$$\beta_1: \quad \hat{\beta}_1 \pm t^* \frac{s}{\sqrt{S_{xx}}} \qquad \beta_0: \quad \hat{\beta}_0 \pm t^* s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

The interval widths reflect estimation uncertainty: more residual noise widens them, while more spread in the predictor values narrows the slope interval.

A Brief Introduction to Multivariate Normality

A multivariate Normal distribution is the vector-valued version of the familiar Normal distribution. It is described by a mean vector and a covariance matrix; every linear combination of its components is Normally distributed.
In regression, collect the errors for all n observations into one vector. Under independent Normal errors with common variance,

$$\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

The covariance matrix $\sigma^2 \mathbf{I}_n$ says each error has variance σ^2 and different errors are independent.
Therefore, conditional on the fixed design matrix $\mathbf{X}$,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad \Longrightarrow \quad \mathbf{y} \mid \mathbf{X} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n).$$

Distributions of Multiple Regression Estimates

Let p be the number of predictors and $q = p + 1$ include the intercept. If $\mathbf{X}$ has full column rank, OLS is a linear transformation of the multivariate Normal response, so

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} \sim \mathcal{N}_q\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right).$$

The off-diagonal entries of this covariance matrix capture dependence between coefficient estimates.
Estimate the noise variance with $s^2 = SS_{res}/(n-q)$, where $n - q = n - p - 1$. For coefficient j, let $c_{jj} = [(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$. Then

$$\frac{\hat{\beta}_j - \beta_j}{s\sqrt{c_{jj}}} \sim t_{n-p-1}, \qquad \hat{\beta}_j \pm t^* s\sqrt{c_{jj}}$$

is the corresponding two-sided confidence interval, with t^* the $1 - \alpha/2$ quantile of t_{n-p-1}. These are individual coefficient intervals.

Partitioning the Variance: The Core Identity

Before we can evaluate our model, we need to understand how OLS physically splits the total variance of our data into two distinct parts. We can write:

$$(y_i - \bar{y}) = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)$$

- $(\hat{y}_i - \bar{y})$: The deviation **explained** by our regression line.
- $(y_i - \hat{y}_i)$: The residual error **unexplained** by our line.

If we square both sides and sum over all n observations, the cross-product cancels out to exactly zero (a fundamental property of OLS residuals). We are left with the fundamental identity of regression:

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

$$\mathbf{SS_{tot}} = \mathbf{SS_{reg}} + \mathbf{SS_{res}}$$

The Cross-Product Term

Let's prove why $SS_{tot} = SS_{reg} + SS_{res}$. We start by expanding the square from our previous identity:

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}\left[(\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)\right]^2$$

$$\sum(y_i - \bar{y})^2 = \sum(\hat{y}_i - \bar{y})^2 + \sum(y_i - \hat{y}_i)^2 + \mathbf{2}\sum(\hat{\mathbf{y}}_\mathbf{i} - \bar{\mathbf{y}})(\mathbf{y}_\mathbf{i} - \hat{\mathbf{y}}_\mathbf{i})$$

To prove our variance cleanly partitions into two pieces, we must prove that the cross-product term at the end is exactly equal to zero.

Let's rewrite it using our residual notation ($e_i = y_i - \hat{y}_i$):

$$\text{Cross-Term} = 2\sum_{i=1}^{n}(\hat{y}_i - \bar{y})e_i = 2\left(\sum_{i=1}^{n}\hat{y}_i e_i - \bar{y}\sum_{i=1}^{n}e_i\right)$$

The Properties of OLS Residuals

1. The Sum of Residuals is Zero

Our derivative with respect to β_0 was: $\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$.
Since $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i) = 0 \implies \sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{e_i} = \mathbf{0}$$

2. Residuals are Orthogonal to Predictors

Our derivative with respect to β_1 was: $\sum x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$.
This translates perfectly to:

$$\sum_{i=1}^{n} x_i(y_i - \hat{y}_i) = 0 \implies \sum_{\mathbf{i=1}}^{\mathbf{n}} \mathbf{x_i e_i} = \mathbf{0}$$

The Cross-product Term

Let's bring our properties back to the expanded cross-term:

$$\text{Cross-Term} = 2\left(\sum_{i=1}^{n} \hat{y}_i e_i - \bar{y}\sum_{i=1}^{n} e_i\right)$$

Because the sum of OLS residuals is zero ($\sum e_i = 0$), the second half vanishes: $\bar{y}(0) = 0$.

$$\text{Cross-Term} = 2\sum_{i=1}^{n} \hat{y}_i e_i$$

We substitute our line equation $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ into the remaining sum:

$$\sum_{i=1}^{n}(\hat{\beta}_0 + \hat{\beta}_1 x_i)e_i = \hat{\beta}_0 \sum_{i=1}^{n} e_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i e_i$$

Using our two properties, this becomes $\hat{\beta}_0(0) + \hat{\beta}_1(0) = 0$.
Conclusion: The cross-term is strictly zero. The variance perfectly partitions!

Analysis of Variance (ANOVA) for Regression

Recall our geometric breakdown of variance: $SS_{tot} = SS_{reg} + SS_{res}$. ANOVA formalizes this by assigning degrees of freedom to each source of variance to calculate the **Mean Squares**.

Source	**Sum of Squares (SS)**	**df**	**Mean Square (MS)**
Regression	$SS_{reg} = \sum(\hat{y}_i - \bar{y})^2$	1	$MS_{reg} = \frac{SS_{reg}}{1}$
Residual	$SS_{res} = \sum(y_i - \hat{y}_i)^2$	$n-2$	$MS_{res} = \frac{SS_{res}}{n-2}$
Total	$SS_{tot} = \sum(y_i - \bar{y})^2$	$n-1$	

Interpreting the Mean Squares

- MS_{res} is simply our unbiased estimate of the noise variance ($\hat{\sigma}^2$).
- MS_{reg} represents the variance successfully captured by the slope of our model.

Evaluating the Fit: R-Squared (R^2)

The ANOVA table gives us the tools to measure how well our line actually fits the data.

The Coefficient of Determination (R^2)

R^2 measures the **proportion of total variance** in our response variable Y that is successfully explained by our regression model.

$$R^2 = \frac{SS_{reg}}{SS_{tot}} = 1 - \frac{SS_{res}}{SS_{tot}}$$

Interpreting R^2

Because $SS_{tot} = SS_{reg} + SS_{res}$, R^2 strictly bounds between 0 and 1:

- $\mathbf{R^2 = 0}$: The model explains nothing. $SS_{reg} = 0$, meaning our line is perfectly flat. We are no better off than just guessing the average ($\bar{y}$).
- $\mathbf{R^2 = 1}$: The model explains everything. $SS_{res} = 0$, meaning there is zero error and every single data point falls perfectly on the line.

Overfitting: When a Model Learns the Noise

Adding predictors gives a regression model more flexibility to fit the observed sample.

- In ordinary least squares, adding predictors cannot increase SS_{res}, so the in-sample R^2 cannot decrease.
- Even a predictor with no real relationship to the outcome may appear useful by chance in a particular sample.
- A model can then fit random fluctuations in the training data rather than the underlying relationship. This is **overfitting**.

A very high training R^2 does not guarantee good predictions for new observations. We therefore want a measure that accounts for model complexity, and we should assess prediction on held-out data when prediction is the goal.

Penalizing Complexity: Adjusted R^2

Ordinary R^2 does not penalize adding predictors. Adjusted R^2 accounts for the degrees of freedom used by the model. Let p be the number of predictors, excluding the intercept:

$$R^2_{\text{adj}} = 1 - \frac{SS_{res}/(n-p-1)}{SS_{tot}/(n-1)} = 1 - (1-R^2)\frac{n-1}{n-p-1}.$$

- The residual mean square is compared with the total mean square, incorporating the number of fitted coefficients.
- Adjusted R^2 can go down when a new predictor adds too little explanatory value to justify its complexity.

Adjusted R^2 helps compare models, but it does not guarantee protection against overfitting; use validation data or cross-validation when evaluating predictive performance.

The F-Test for Overall Significance

We want to test if our model actually explains anything at all.

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 \neq 0$$

The Signal-to-Noise Ratio

We construct a test statistic by comparing the variance our model explains (the signal) against the variance it couldn't explain (the noise):

$$F = \frac{MS_{reg}}{MS_{res}} = \frac{\text{Explained Variance}}{\text{Unexplained Variance}} \sim F_{1,n-2}$$

- If H_0 is true (the model is useless), F will be close to 1.
- If H_0 is false (the model is good), the signal overwhelms the noise, and F becomes very large.

For a single predictor, the F-test and the t-test are identical!

$$F_{1,n-2} = (t_{n-2})^2$$

A Few Properties of the F Distribution

An F random variable is a ratio of two independent chi-square variables, each divided by its degrees of freedom:

$$F = \frac{U/d_1}{V/d_2}, \qquad U \sim \chi^2_{d_1}, \quad V \sim \chi^2_{d_2}, \quad U \perp V.$$

- F is always nonnegative and typically right-skewed; its shape depends on both degrees of freedom.
- Swapping the degrees of freedom reciprocates the distribution: if $F \sim F_{d_1,d_2}$, then $1/F \sim F_{d_2,d_1}$.
- When $d_2 > 2$, $E[F] = d_2/(d_2 - 2) > 1$; when $d_2 \leq 2$, the mean is undefined.

For an overall regression test, evidence against H_0 lies in the **right tail**: large ratios of explained to unexplained variation are unlikely under the null. In simple regression, if $T \sim t_{n-2}$ then $T^2 \sim F_{1,n-2}$.

Beyond Linear Regression: The GLM Framework

Ordinary linear regression is not ideal for every outcome. A **generalized linear model (GLM)** keeps a linear predictor but allows other response distributions and a link function.

1. **Response distribution:** Specify the distribution of $Y_i \mid \mathbf{x}_i$ (for example, Normal, Bernoulli, or Poisson).
2. **Linear predictor:** Combine predictors linearly:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

3. **Link function:** Relate the conditional mean $\mu_i = E[Y_i \mid \mathbf{x}_i]$ to the linear predictor:

$$g(\mu_i) = \eta_i.$$

The link transforms the **mean response**, not each observed outcome. With a Normal response and identity link $g(\mu) = \mu$, this reduces to ordinary linear regression.

Normal Response: Identity Link

For predictors $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$, define

$$\eta_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}.$$

The Normal-response model is

$$Y_i \mid \mathbf{x}_i \sim \mathcal{N}(\mu_i, \sigma^2), \qquad \mu_i = E[Y_i \mid \mathbf{x}_i].$$

The identity link sets $g(\mu_i) = \mu_i = \eta_i$. With independent Normal errors of common variance, this is the standard linear regression model, and maximizing its likelihood gives the OLS coefficient estimates. Each β_j is the additive mean change per one-unit increase in x_{ij}, holding other predictors fixed.

What Is a Canonical Link?

Many response distributions can be written in exponential-family form:

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}.$$

The parameter θ multiplying y in the log-likelihood is called the **natural parameter**. The mean is $\mu = b'(\theta)$, so the canonical link is the function that maps the mean to this natural parameter:

$$g(\mu) = (b')^{-1}(\mu) = \theta.$$

A GLM with the canonical link sets $g(\mu_i) = \eta_i = \mathbf{x}_i^T\boldsymbol{\beta}$. This choice follows naturally from the response distribution; it is common, but not the only possible link. The Poisson and Bernoulli examples show how their canonical links arise from their probability models.

Poisson Counts: Why the Log Link?

For a count $Y_i \in \{0, 1, 2, \ldots\}$, assume $Y_i \mid \mathbf{x}_i \sim \text{Poisson}(\mu_i)$, with probability mass function

$$\begin{aligned}\Pr(Y_i = y_i \mid \mathbf{x}_i) &= \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!} \\ &= \exp\{y_i \log \mu_i - \mu_i - \log(y_i!)\}.\end{aligned}$$

The coefficient of y_i in the exponent is the natural parameter $\theta_i = \log \mu_i$. A GLM uses this as its linear predictor, $\theta_i = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$, giving the **canonical log link**:

$$g(\mu_i) = \log \mu_i = \eta_i, \qquad \mu_i = e^{\eta_i} > 0.$$

Thus, a one-unit increase in x_{ij} multiplies the expected count by e^{β_j}, holding other predictors fixed.

Binary Outcomes: Why the Logit Link?

For $Y_i \in \{0, 1\}$, assume $Y_i \mid \mathbf{x}_i \sim \text{Bernoulli}(p_i)$, where $p_i = \Pr(Y_i = 1 \mid \mathbf{x}_i)$:

$$\begin{aligned}\Pr(Y_i = y_i \mid \mathbf{x}_i) &= p_i^{y_i}(1 - p_i)^{1-y_i} \\ &= \exp\left\{ y_i \log\left(\frac{p_i}{1 - p_i}\right) + \log(1 - p_i) \right\}.\end{aligned}$$

The coefficient of y_i in the exponent is the natural parameter $\theta_i = \log\left(\frac{p_i}{1-p_i}\right)$. Setting $\theta_i = \eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ gives the **canonical logit link**:

$$g(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \eta_i, \qquad p_i = \frac{1}{1 + e^{-\eta_i}}.$$

Substituting $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ into the Bernoulli log-likelihood gives

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[y_i \mathbf{x}_i^T \boldsymbol{\beta} - \log\Big(1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta})\Big) \Big].$$

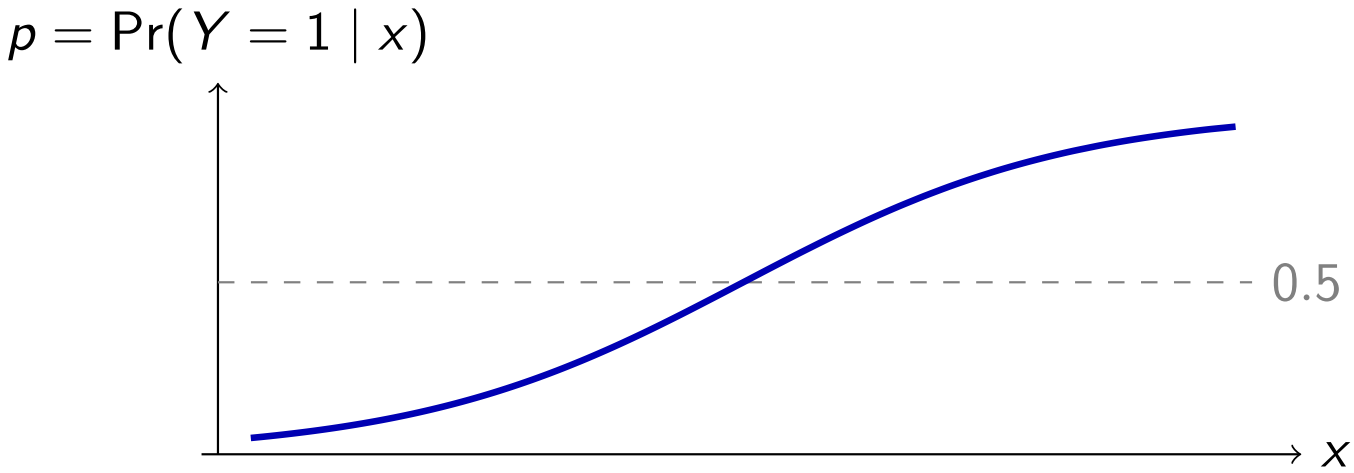
The inverse link keeps probabilities between 0 and 1.

Logistic Regression Models Probabilities

For a binary response, logistic regression models the probability of the outcome $Y_i = 1$:

$$\text{logit}(p_i) = \beta_0 + \beta_1 x_i, \qquad p_i = \Pr(Y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}.$$

The linear predictor can take any real value, while the inverse-logit maps it to a probability between 0 and 1.



The S-shaped curve shown uses $\beta_0 = 0$ and $\beta_1 = 1$; changing the coefficients shifts or changes its steepness.

Interpreting Logistic Regression Coefficients

For one predictor, β_1 is the change in **log-odds** for a one-unit increase in x. Exponentiating gives the odds ratio:

$$\frac{\text{odds}(x+1)}{\text{odds}(x)} = e^{\beta_1}, \qquad \text{odds}(x) = \frac{p(x)}{1-p(x)}.$$

Example: $\beta_1 = \log 2$

The odds ratio is $e^{\beta_1} = 2$: each one-unit increase in x multiplies the odds by 2, a 100% increase. If $\beta_0 = \log(0.25)$, then:

x	0	1	2
odds(x)	0.25	0.5	1
$p(x)$	0.20	1/3	0.50

Odds of 0.25 mean 1:4 odds and probability 0.20; odds of 0.5 mean 1:2 odds and probability 1/3. The odds double, but the probability does not.

An odds ratio above 1 raises odds; below 1 lowers them; 1 means no change. Probability changes depend on baseline probability. In multiple-predictor models, e^{β_j} is the per-unit odds multiplier, holding other predictors fixed.

How Logistic Regression Is Fit

Assuming independent Bernoulli outcomes, each observation contributes

$$\Pr(Y_i = y_i \mid \mathbf{x}_i) = p_i^{y_i}(1-p_i)^{1-y_i}, \qquad p_i = \frac{1}{1+e^{-\mathbf{x}_i^T\beta}}.$$

The likelihood is

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{1-y_i}.$$

Its log-likelihood, written directly in terms of the predictors and coefficients, is

$$\ell(\beta) = \sum_{i=1}^{n}\Big[y_i\mathbf{x}_i^T\beta - \log\Big(1+\exp(\mathbf{x}_i^T\beta)\Big)\Big].$$

We choose $\hat{\beta}$ to maximize this log-likelihood (equivalently, minimize the cross-entropy loss). Unlike ordinary least squares, there is generally no closed-form solution, so software uses an iterative numerical method.
The fitted model first gives probabilities. Turning them into class labels requires a separate decision threshold, chosen for the application.

From Predicted Probabilities to Decisions

A fitted logistic regression model returns an estimated probability $\hat{p}_i = \Pr(Y_i = 1 \mid \mathbf{x}_i)$, not a class label. To make a decision, choose a threshold t:

$$\widehat{Y}_i = \begin{cases} 1, & \hat{p}_i \geq t, \\ 0, & \hat{p}_i < t. \end{cases}$$

The threshold is a decision rule chosen after fitting; it is not another regression coefficient. The default $t = 0.5$ is convenient, but not automatically best. Lowering t labels more cases positive, typically catching more true positives while also increasing false positives. The preferred balance depends on the consequences of each type of mistake.

Evaluating Decisions: The Confusion Matrix

Compare predicted labels with known outcomes on validation data:

	Predicted positive	Predicted negative
Actually positive	TP	FN
Actually negative	FP	TN

$$\text{Sensitivity (recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}},$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Accuracy} = \frac{\text{TP} + \text{TN}}{n}.$$

Sensitivity measures how many actual positives are found; precision measures how many positive predictions are correct. Accuracy alone can be misleading when one class is rare.

Choosing a Decision Threshold

If false positives cost C_{FP} and false negatives cost C_{FN}, then with calibrated probabilities and only these two error costs, predict positive when

$$\hat{p}_i > t^*, \qquad t^* = \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}.$$

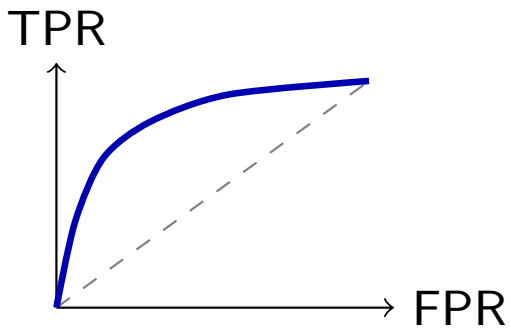
A high cost for missed positives lowers the threshold; a high cost for false alarms raises it. With equal costs, this decision rule gives $t^* = 0.5$.
In practice, compare candidate thresholds on validation data. ROC curves show sensitivity versus false-positive rate; precision–recall curves are especially informative when positives are rare. Choose the threshold for the application, then report final performance on a separate, untouched test set.

ROC Curves: Sensitivity and False Positives

At each threshold, calculate the true-positive rate (TPR) and false-positive rate (FPR):

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}.$$

The ROC curve plots TPR against FPR; curves nearer the upper-left indicate better ranking.



The area under the ROC curve (AUC) is

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(u)\, du, \qquad u = \mathrm{FPR},$$

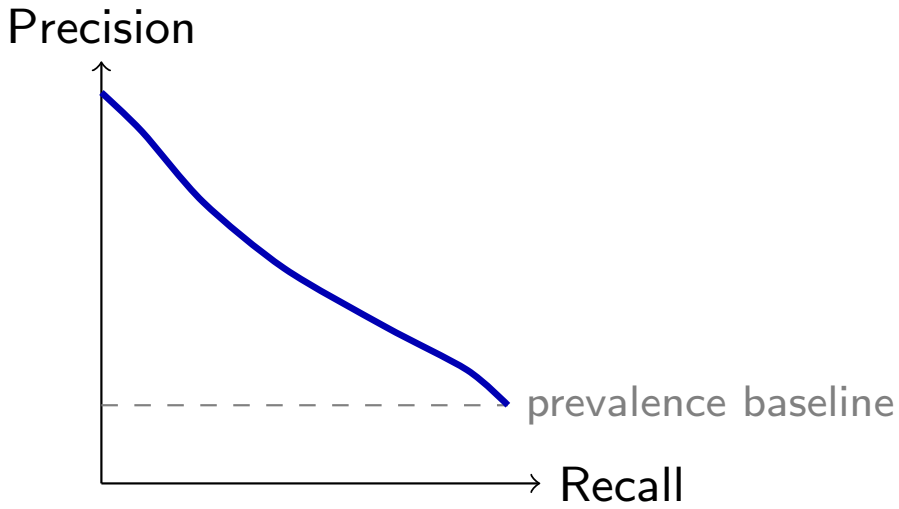
or the probability a random positive ranks above a random negative. AUC measures ranking, not calibration or threshold choice.

Precision–Recall Curves

A precision–recall (PR) curve plots precision against recall as the probability threshold varies:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

Raising the threshold often increases precision but reduces recall: fewer cases are labeled positive, so fewer false alarms may occur, but more true positives may be missed.



PR curves are especially useful when positives are rare; the dashed no-skill baseline equals their prevalence. Compare curves on the same evaluation population, select a threshold using validation data, and assess it on held-out test data.

When Is Logistic Regression the Wrong Tool?

Standard logistic regression models a single binary outcome. Choose a model that matches the response:

- **Continuous outcome:** use a continuous-response model such as linear regression.
- **Count outcome:** consider a Poisson or negative-binomial model.
- **Several outcome categories:** use multinomial or ordinal logistic regression, depending on whether categories are ordered.
- **Time until an event:** use survival methods to account for censoring.

The standard model also assumes independent observations. Repeated measurements or clustered data may require a mixed-effects model or another method that accounts for dependence.

When Standard Logistic Regression Struggles

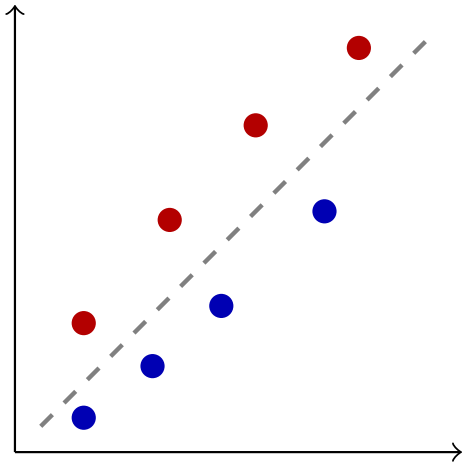
Complete Separation

Example: two biomarkers perfectly distinguish patients with and without the disease, as shown. Let $\eta_i = \beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}$. Perfect separation means $(2y_i - 1)\eta_i > 0$ for every observation. Scale all coefficients by $c > 0$:

$$\hat{p}_i(c) = \frac{1}{1 + e^{-c\eta_i}}.$$

$$\ell(c\beta_0, c\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i c\eta_i - \log(1 + e^{c\eta_i}) \right] \longrightarrow 0, \quad c \to \infty.$$

The coefficient magnitude diverges, but $L \to 1$ (its upper bound), so $\ell = \log L \to 0$. No finite coefficients attain this supremum. Curvature, quasi-separation, sparse data, and collinearity may also cause problems; transformations or penalization can help.



Two measurements; blue: controls, red: cases.